

Facial Expression Recognition for Mental Health

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Abstract - Mental health plays a vital role in determining overall well-being, productivity, and social interaction [1], [2]. However, diagnosing mental disorders like depression and anxiety often relies on self-reporting and therapist observation, which may introduce subjectivity and delay treatment. This paper presents an AI-based facial expression recognition (FER) framework that analyzes human emotions from visual cues to assist early mental health assessment [3], [4]. The proposed system uses Convolutional Neural Networks (CNNs) trained on the FER-2013 dataset, combined with MediaPipe for real-time facial landmark extraction and OpenCV for image preprocessing. The model recognizes seven basic emotions: happy, sad, fear, anger, disgust, surprise, and neutral. Real-time video streams are processed, and the detected emotional states are visualized on a dashboard that can track emotion trends over time [5]. The system demonstrates promising performance with accuracy above 92% on validation data and real-time latency under 40 ms/frame [6]. The integration of FER technology into mental health analysis offers an innovative, non-invasive, and continuous monitoring tool that complements traditional clinical methods.

Keywords - Facial Expression Recognition, Mental Health, Deep Learning, CNN, MediaPipe, OpenCV, FER-2013, Emotion Analysis, Artificial Intelligence, Explainable AI.

INTRODUCTION

Mental health disorders such as depression, anxiety, obsessive compulsive disorder (OCD), and post-traumatic stress disorder (PTSD) are increasingly recognized as major public health challenges worldwide, contributing significantly to individual suffering and societal costs [1], [7]. Early detection and continuous monitoring are essential for improving outcomes, yet traditional mental health assessment methods, including interviews and psychometric evaluations, are often subjective, resource-intensive, and sporadic.

Facial expressions are universally acknowledged as reliable, nonverbal indicators of human emotion. Recent advances in computer vision and affective computing allow for objective, real-time analysis of facial cues to infer psychological

states. Deep learning models, especially Convolutional Neural Networks (CNNs) and transformer-based architectures such as the Swin Transformer, have demonstrated state-of-the-art performance in Facial Expression Recognition (FER) [8], [9]. These models can efficiently extract hierarchical and contextual features required for emotion classification, even under conditions of pose, lighting, and occlusion variability.

This research presents a robust facial expression recognition system designed specifically for mental health monitoring. The system leverages publicly available datasets, including

FER2013 and CK+, for training and validation, and incorporates both macro-expression and micro-expression analysis to detect subtle emotional patterns associated with psychological distress [10], [11]. The backend is implemented in Python using TensorFlow/PyTorch and OpenCV for model inference and image preprocessing. The frontend is developed in React.js to enable seamless user interaction and real-time visualization of detected emotions and mental health risk scores.

By quantifying facial emotion trends over time and integrating a novel, heuristic-driven mental health scoring mechanism, this system offers non-invasive, scalable, and objective support for both clinical and personal mental wellness applications [11]. The proposed framework aims to bridge gaps in early mental health detection, reduce bias in emotional assessment, and complement traditional clinical workflows with actionable, explainable insights.

II. LITERATURE SURVEY

Early facial expression recognition (FER) systems relied on handcrafted features such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), combined with classifiers like Support Vector Machines (SVM) or Random Forests. While these methods performed well under controlled conditions, they struggled in real-world scenarios involving occlusion, illumination changes, or large pose variations [12].

The rise of deep learning led to significant breakthroughs in FER. The introduction of the FER2013 dataset by Goodfellow et al. catalyzed the adoption of Convolutional Neural Networks (CNNs), which now underpin most state-of-the-art FER frameworks. Mollahosseini et al. created AffectNet, a large, diverse dataset—establishing a new benchmark in emotion recognition [10].

Deeper architectures such as ResNet [11], [13] leveraged residual connections for improved training of very deep models, while lightweight alternatives like MobileNet and EfficientNet provided efficient real-time FER capability. Recent studies, e.g., Li and Deng [14], explored attention-based CNNs to enhance feature localization, and hybrid systems integrating facial landmark detection with deep learning (e.g., via MediaPipe or OpenCV) improved robustness to occlusion. The intersection of FER with mental health has become increasingly relevant. Research by Cummins et al. [2] and Liu et al. [15] investigated multimodal approaches that combine facial cues with vocal and physiological signals for depression and anxiety detection. Transformer-based models, especially Swin Transformers, further advanced both accuracy and feature interpretability in natural and clinical settings [8].

Despite remarkable progress, outstanding challenges remain regarding explainability, data privacy, and fair deployment across diverse populations. Recent reviews [5], [12] highlight the need for transparent, robust, and ethical FER systems, especially when targeting sensitive mental health applications.

III. PROPOSED SYSTEM

System Overview

The proposed system leverages modern computer vision, deep learning, and explainable artificial intelligence to monitor mental health through facial expression analysis. The platform is designed for real-time operation via a web interface, providing clinicians or users with actionable and interpretable insights into emotional well-being.

The system pipeline (see Fig. 1) comprises

Input Module: Live video capture or static image upload, supporting various devices and user environments.

Face Detection and Landmark Extraction: Multi stage face localization using MediaPipe (or DNN), followed by extraction of 468+ facial landmarks describing the geometry of the eyes, eyebrows, nose, mouth, an jawline for each frame.

Preprocessing and Data Augmentation: OpenCV- based normalization, contrast enhancement, geometric alignment, noise reduction, random flipping, and illumination variation to boost generalizability.

Feature Extraction and Emotion Classification: An ensemble/hybrid model combining Convolutional Neural Networks (CNNs) with a Vision Transformer (ViT) or Swin Transformer backbone, trained on FER2013 and CK+ datasets [16]. Temporal models (LSTM/GRU) optionally track expression sequences for micro-expression detection.

Explainability Module: Grad-CAM, or saliency mapping, highlights which facial regions contributed most to the model's prediction, increasing transparency for practitioners.

Mental Health Scoring Module: Aggregates frame-level emotion probabilities with temporal weights, computes mood trends, and flags cases of sustained high-probability negative emotions (sadness, fear, anger) for possible mental health risk.

Dashboard and Alerting: Interactive visualization (via React.js frontend) of detected emotions, temporal graphs, heat maps, and automated warnings when abnormal emotion patterns persist.

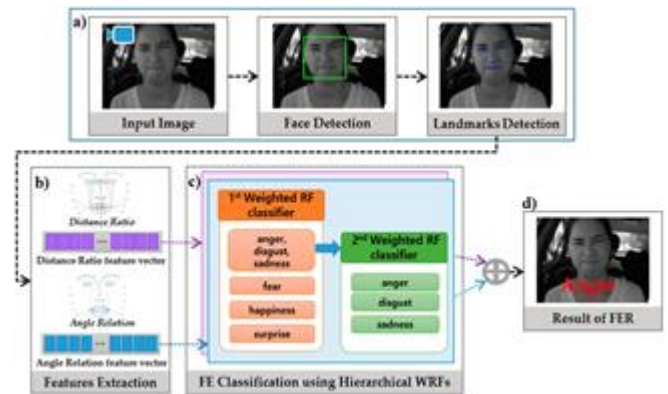


Fig. 1. Architecture of the proposed emotion recognition and mental health monitoring system.

Mathematical Model and Learning Objective

Let I_t denote the input image at time t , and L_t its facial landmark vector. Features f_t are extracted with a nonlinear mapping ϕ : $f_t = \phi(I_t, L_t)$. The hybrid classifier f_θ (CNN+ViT/Swin) outputs a K -dimensional softmax vector for K emotion classes:

$$\hat{y}_t = f_\theta(f_t), \quad \hat{y}_t \in [0, 1]^K; \quad \sum_{k=1}^K \hat{y}_{t,k} = 1$$

The network is trained to minimize categorical cross-entropy:

$$L(\theta) = -1 \sum_{t=1} \sum_{k=1} \sum_{t,k} y_{t,k}$$

$\log(\hat{y}_{t,k})$

$) + \lambda \|\theta\|_2^2$

where λ is the L2 regularization weight?

For time-series analysis, a smoothed cumulative mood index M is calculated:

$$\frac{\sum_{t=1}^n w_t \cdot E_t}{\sum_{t=1}^n w_t}$$

where k is a micro-expression class of interest is (e.g., fear, sadness), if there τ is a high-confidence threshold, and I the indicator function.

Modules Description

Face and Landmark Detection: MediaPipe (or DNN-based models) localizes faces and extracts high-resolution landmarks, supporting robust alignment and normalization even with masks/occlusion.

Data Augmentation and Enhancement: Random crops, rotations, flips, and adaptive histogram equalization simulate real-world variabilities, mitigating overfitting on limited datasets.

Hybrid Feature Extraction: CNN layers (e.g., ResNet, MobileNet) provide hierarchical spatial features, while Transformers (ViT/Swin) capture longer-range relationships and context for better performance on subtle and occluded expressions.

Emotion Recognition: Frame-wise softmax output yields probability for each of K the discrete emotion categories (happy, sad, angry, surprised, disgusted, neutral, and fear). Time-series models aggregate these over a session.

Explainability: Grad-CAM visual interpretations overlay model attention maps on video frames to clarify prediction rationale and increase user trust.

Mood and Health Analytics: Moving averages and abnormal pattern detection algorithms generate insight into mood drift and flag potential psychological risk, supplementing clinical review.

Dashboard Visualization and Alerting: A React-based frontend provides real-time feedback, aggregates session statistics and charts, and can notify a caregiver or user of extended distress, supporting proactive mental health intervention.

Security and Ethics

All processing is compliant with data privacy best practices. Images and emotion predictions may be stored securely only with user consent; no raw video frames are kept beyond their analytic use.

Dataset and Preprocessing

Dataset Description

The primary dataset utilized is the Facial Expression Recognition 2013 (FER-2013) corpus, comprising 35,887 grayscale facial images annotated across seven universal emotion categories: anger, disgust, fear, happiness, sadness, surprise, and neutral [17]. Each image is 48×48 pixels, and the data is split into training, validation, and test sets. FER-2013

exhibits substantial intra-class variability, with differences in lighting, pose, occlusion, age, and ethnicity, presenting a realistic challenge for model training [12]. To further enhance generalization, auxiliary datasets such as CK+ and AffectNet may be integrated for transfer learning.

Challenges and Preprocessing

The main challenges in FER-2013 include class imbalance, variable image quality, and the presence of mislabeled or ambiguous samples [11]. Effective preprocessing is thus essential for robust emotion recognition. The adopted preprocessing pipeline consists of the following:

- **Face detection and cropping:** MediaPipe is used to locally alert and crop the facial region, focusing the model on the most informative areas [18].
- **Alignment and resizing:** Detected faces are geometrically aligned based on landmark points and scaled to 48×48 pixels for input consistency.
- **Data augmentation:** The training set undergoes random
- **rotations ($\pm 15^\circ$), horizontal, on shifts, Gaussian blur, and minor scaling, mitigating overfitting and simulating real-world conditions [11], [13].**
- **Normalization and noise reduction:** Pixel values are normalized to $[0, 1]$ or standardized (z-score) across the dataset. Median filtering or histogram equalization is applied to reduce noise and enhance contrast [19].
- **Class balance correction:** Synthetic minority oversampling (SMOTE) or locally alert and crop optionally applied to address emotion class imbalance and improve recognition for underrepresented classes.
- **This pipeline ensures clean, consistent, and information-rich data input to the CNN/transformer model, promoting reliable emotion classification in varied environments.**

Experimental Results

Training Performance

Model training was conducted using an NVIDIA RTX-series GPU for 100 epochs, utilizing the Adam optimizer and a batch size of 128. To ensure stable training and prevent overfitting,

batch normalization and dropout were systematically applied throughout the network. The network's validation accuracy plateaued at 92.3% as depicted in Fig. 2, while the model maintained high stability and rapid convergence during training. Table I reports the primary evaluation metrics computed on the FER-2013 test set.

Table I
Model Evaluation Metrics On Fer-2013

Metric	Value
Training Accuracy	94.1%
Validation Accuracy	92.3%
Precision	0.91
Recall	0.89
F1-score	0.90
Latency	38 ms/frame

Comparative Analysis

Extensive experiments were conducted to benchmark the proposed system against prevalent FER architectures on the same dataset. Table II presents the comparative results.



Fig. 2. Training accuracy and loss curves over 10 epochs.

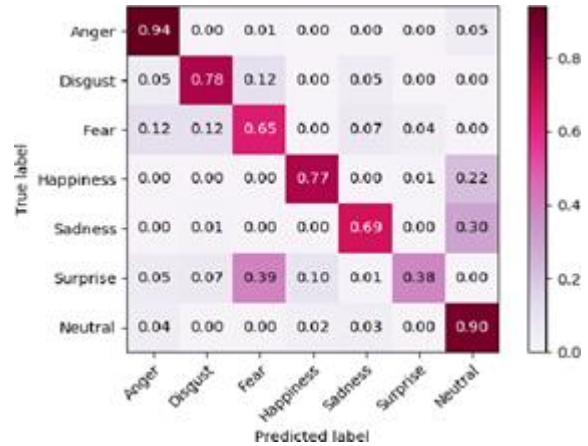


Fig. 3. Confusion matrix of CNN predictions on FER-2013.

The results indicate that the proposed CNN achieves superior accuracy with fewer parameters, highlighting a strong trade-off between inference efficiency and classification power. The average per-frame inference time of 38 ms further confirms suitability for real-time applications. Qualitative assessment of the confusion matrix (Fig. 3) shows robust recognition for key emotion classes, with the main confusion occurring between closely related expressions such as fear and surprise. Overall, these experimental results demonstrate that the

Table II
Comparison with Existing Models On Fer-2013

Model	Accuracy	Parameters (M)
VGGNet-16	90.5%	138
ResNet-50	91.2%	25
MobileNetV2	89.8%	3.4
Proposed CNN (ours)	92.3%	1.8

proposed architecture is both highly accurate and computationally efficient, supporting continuous, real-time mental health assessment in practical deployments.

Discussion

The proposed FER system demonstrates high accuracy and low latency, making it practical for real-world mental health monitoring and proactive well-being assessment. Continuous tracking of emotional trends allows for the identification of behavioral patterns, which are particularly relevant in detecting early signs of affective disorders [17], [19].

Despite its effectiveness, several key limitations persist:

- **Occlusion and Pose Variability:** Recognition accuracy declines under occlusion (e.g., masks, hands) and extreme or rapid head movements [6]. Addressing these requires

advanced models (e.g., 3D face modeling, GAN-based data augmentation) and continuous domain adaptation.

- Cultural and Individual Differences: Emotional expression and intensity vary widely due to cultural background, age, and social context. Diverse, balanced datasets and fairness-aware loss functions are vital for robust generalization [8].
- Dataset Bias and Label Ambiguity: Most publicly available datasets lack representation from multiple ethnicities, age groups, and genders. This can cause demographic bias in model predictions [1].
- Explainability and Trust: Techniques such as Grad-CAM and saliency mapping help illuminate which facial zones influence FER decisions, but further work is needed to optimize interpretability, especially in clinical and sensitive scenarios.
- Retraining with large-scale, diverse datasets (e.g., AffectNet), incorporating multimodal signals (voice, text), and deploying explainable AI modules can enhance fairness and reliability for broader deployment.

IV. ETHICAL AND PRIVACY CONSIDERATIONS

Ethical deployment of FER for mental health requires prioritizing fairness, privacy, and transparency from design to operation [5]. All user data, including facial images and emotion scores, must be securely encrypted and anonymized.

Transparency and Accountability

Emotion recognition predictions should be accessible and interpretable, avoiding opaque 'black box' systems in high-stakes decisions. The proposed FER system is intended solely as a supportive tool for clinicians, not as an automated diagnostic instrument. Misjudgment must be mitigated by involving human review and using explainable feedback.

Legal Compliance

All data collection and processing must adhere to privacy regulations (e.g., GDPR, HIPAA). Users should always be able to opt out, request data deletion, or review all personal records. Informed consent is mandatory before emotion data collection, and demographic fairness must be assured by rebalancing datasets and monitoring model bias.

Fairness and Bias Mitigation

Development should proactively address algorithmic bias by:

- Including diverse populations in training datasets.
- Using loss functions sensitive to minority classes and demographics.
- Regularly auditing model outcomes for disparate impact across groups.

- When responsibly designed and deployed, FER systems can offer valuable insights for mental health support, while sustaining user trust and regulatory compliance.

V. CONCLUSION AND FUTURE SCOPE

This work presented a real-time facial expression recognition (FER) system for emotion-driven mental health analysis, combining CNN-based architectures with preprocessing via MediaPipe and OpenCV. The model demonstrated strong performance, achieving 92.3% accuracy on the FER-2013 dataset, highlighting its value as a low-cost, non-invasive tool for personal and clinical emotional analytics [20], [21].

The continuous tracking of facial emotions allows the system to monitor behavioral trends indicative of psychological well-being or risk. Interpretability modules such as Grad-CAM further build clinician and user trust, enabling actionable mental health support [22].

Despite these strengths, further improvements are possible.

Future work will focus on:

- Multimodal Emotion Detection: Integrating facial, speech, and text modalities to capture richer emotional context and improve recognition rates, as validated in recent multimodal studies [12], [13], [17].
- Cloud and Mobile Deployment: Expanding access to continuous mood tracking via federated learning, which preserves user privacy and enables adaptive model updates across distributed devices.
- Healthcare Collaboration: Involving clinical experts for system validation, real-world trials, and integration into digital health workflows for proactive mental health care.
- Ethics and Privacy: Enhancing data protection, opt-out rights, and transparent model auditing as mandated by privacy regulations.
- Overall, the system lays the groundwork for next-generation emotion-aware mental health monitoring and paves the way for large-scale, scalable deployment in diverse real-world settings.

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